A LOCAL OBSTRUCTION PROOF OF THE RIEMANN HYPOTHESIS

JOHN SHIELDS

Abstract. We present a local-geometric obstruction proof of the Riemann hypothesis. The two-by-two jet-normalized blocks of the phase kernel of ζ at one or two centers form a finite-dimensional comparison object: a hypothetical off-line zero forces a transverse mode that the actual zeta phase, after a single fixed value-gauge quotient, suppresses at a strictly larger polynomial exponent in the local scale. Two-point and four-point comparisons of this object aggregate over heights via N-point odd-moment cancellation to yield an unconditional zero-free strip around the critical line, hence the Riemann hypothesis. A self-contained final section provides an optional source-energy supply available to the two-point and four-point comparisons if their standalone forms prove insufficient.

Contents

Date: May 2026.

2020 *Mathematics Subject Classification.* 11M26 (primary), 11M06, 11M50 (secondary).

Key words and phrases. Riemann hypothesis, zeta function, critical line, phase kernel, jet normalization, jet-Gram block, odd-moment cancellation, source-energy supply.

Preprint, clean rebuild draft.

1. Introduction

paper: [**LIVE**] v1: [**n/a**] referee: [**not-started**] python: [**n/a**] lean: [**not-started**]

The Riemann hypothesis, originating in Riemann's 1859 memoir [4], asserts that every nontrivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$; see [1, 2, 3, 5] for surveys. The proof presented in this paper proceeds by a local-geometric obstruction. For each height γ we work on a smooth window I_T centered at $T = \gamma$ of width comparable to $1/Q(\gamma)$, where Q is a scale parameter slowly growing with $|\gamma|$. On this window we fix a real local phase Φ for ζ and study the phase kernel

$$K_\Phi(x, y) = \frac{\sin\big(\Phi(x) - \Phi(y)\big)}{\pi(x - y)}, \qquad x \neq y,$$

extended by removable singularity at $x = y$. The 2×2 blocks of K_Φ at one or two centers, transformed into the value/derivative basis by a fixed elementary matrix P_h, become explicit matrices $J(T)$ and $N_{12}(T_1, T_2)$ whose entries are finite combinations of $q = \Phi'$, q', q'', and $\Delta = \Phi(T_1) - \Phi(T_2)$. All subsequent estimates reduce to finite linear-algebraic manipulations of these matrices.

A hypothetical off-line zero $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ forces a specific transverse mode in the value/derivative space — the toy mode — which survives the natural value/gauge quotient. The actual zeta phase, under the same quotient, cannot realize that mode: the transverse channel of the actual whitened block is suppressed at a strictly larger polynomial exponent in Q than the toy lower bound. The clash is two-point at minimum and four-point in the full symmetry quartet $\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$, propagated across heights by an N-point odd-moment cancellation. Aggregation over the dyadic height ladder gives an unconditional zero-free strip around the critical line; combined with standard low-height verification of RH, this yields the Riemann hypothesis.

The argument is self-contained except for one optional input. If either the two-point or the four-point standalone closure proves insufficient, the source-energy lower bound on the canonical source curvature $q''_{\mathrm{can}} = \partial_t^3 Z$, stated as Hypothesis 12.3 in Section 12, supplies the additional polynomial-in-Q room required.

1.1. Active dependency graph.

The active spine of the argument is the chain

$$\begin{aligned} &\S2 \longrightarrow \S3 \longrightarrow \S4 \\ &\longrightarrow \S5,\ \S6,\ \S7 \\ &\longrightarrow \S8 \\ &\longrightarrow \S9 \longrightarrow \S10 \longrightarrow \S11. \end{aligned}$$

Section 12 stands aside: it provides the optional source-energy supply consumed by Section 8 when their standalone forms are insufficient. All cross-references to Section 12 from earlier sections are forward references; that section is self-contained.

1.2. Master conditional theorem.

Theorem 1.1 (Master conditional). *Assume the same-point and cross-block jet-limit identities (Lemmas 3.1 and 3.3); the finite-scale whitening loss bound (Lemma 4.3); the toy anomaly visibility lower bound (Theorem 5.3); the two-point or four-point actual-zeta transverse suppression upper bound (Hypothesis 7.1 or Hypothesis 7.3); the corresponding local rigidity theorem (Theorem 8.1 or Theorem 8.4); the N-point odd-moment cancellation (Lemma 9.1); and the aggregation step (Theorem 10.1). Then every nontrivial zero of ζ satisfies* $\operatorname{Re} s = \frac{1}{2}$.

Gap 1.2. The conditional statement is firm; its inputs are the section-level results listed above. The proof is short — a composition of the section-level results — and is written out in Section 11.

1.3. **Notation and conventions.** We use the symmetric pair convention $T \pm h$ (rather than $T \pm h/2$) throughout. The point-to-jet transform P_h of (2.2) is fixed by this choice. The scale parameter $Q = Q(\gamma)$ is a fixed slowly growing function of the height; all polynomial losses are stated as powers of Q with absolute exponents. "Retained packets" refers to a polynomial-weight sub-family of the local packets at height γ on which the chart-regularity and Gram nondegeneracy hypotheses of Definition 2.1 hold.

2. Local kernel and jet normalization

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

A candidate off-line zero ρ at height γ is detected only at scales comparable to $1/\gamma$, and the analytic data of ζ beyond a window of width $\asymp 1/Q(\gamma)$ around γ does not enter the comparison object. We therefore fix a real smooth phase Φ for ζ on such a window once and for all, and never use ζ again — only Φ, and only through the phase kernel K_Φ and its Taylor jets at one or two centers.

2.1. **Local phase chart.**

Definition 2.1 (Local phase chart)**.** For each height T we fix a real smooth function Φ_T on an interval I_T of length $\asymp 1/Q(T)$ centered at T with the following properties:

(P1) Φ_T is the standard real critical-line phase of ζ on I_T, in a chart whose denominators admit a polynomial-in-Q lower bound;

(P2) the normalized derivative $q := \Phi_T'$ satisfies a polynomial-in-Q lower spectral bound on the retained packets at height T.

We write Φ and q for Φ_T, q when T is fixed by context.

Gap 2.2. Existence of such a chart at every sufficiently high T and the polynomial constants in (P1)–(P2) must be established before Definition 2.1 is invoked downstream. v1 prose to mine: the jet-limit local blocks subsection of the local kernel and jet normalization section.

Definition 2.3 (Phase kernel)**.** The phase kernel of Φ on I_T is

$$K_\Phi(x,y) = \begin{cases} \dfrac{\sin\bigl(\Phi(x)-\Phi(y)\bigr)}{\pi(x-y)}, & x \neq y, \\ \dfrac{q(x)}{\pi}, & x = y. \end{cases} \tag{2.1}$$

2.2. **Point-to-jet transform.** The kernel K_Φ is used only at small symmetric pairs $T \pm h$; the 2×2 block at such a pair is rotated out of the point basis into the value/derivative basis by the fixed matrix

$$P_h = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1/(2h) & 1/(2h) \end{pmatrix}. \tag{2.2}$$

Remark 2.4 (Pair separation convention)**.** We use the symmetric pair $T \pm h$, not $T \pm h/2$; the pair separation is therefore $2h$, and the $1/(2h)$ factor in the lower row of P_h is the inverse separation. All formulas in Sections 3 and 4 are stated under that convention. Any imported lemma using the alternative convention $T \pm h/2$ (or using a P_h with $1/h$ in the lower row) must be retransformed before insertion.

Remark 2.5 (Normalization audit)**.** The orthonormal value/derivative normalization (2.2) is fixed because it is the unique normalization of P_h for which $P_h A_h(T) P_h^\top$ converges to the matrix $J(T)$ of (3.1) entry by entry as $h \to 0$; see Lemma 3.1. Alternative choices — in particular the unnormalized $\frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ -1/h & 1/h \end{smallmatrix}\right)$ appearing in some early drafts — rescale the $(1,1)$ and $(2,2)$ entries of the limit by reciprocal factors of 2 and break the displayed identity. This normalization is therefore fixed for every finite-scale block in Section 4.

3. Jet-limit local blocks

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

The phase kernel K_Φ collapses, in the limit $h \to 0$, into two explicit 2×2 matrices in the value/derivative basis: a same-point block $J(T)$ attached to a single center, and a cross-block $N_{12}(T_1, T_2)$ attached to two centers. These matrices, together with the polynomial Gram lower bounds of Section 3.3, are the only objects manipulated downstream; the phase Φ reappears only through q, q', q'' and $\Delta = \Phi(T_1) - \Phi(T_2)$.

3.1. Same-point jet-limit.

Lemma 3.1 (Same-point jet-limit)**.** *Let $A_h(T)$ denote the 2×2 block of K_Φ at the points $T-h, T+h$. Then*

$$P_h\, A_h(T)\, P_h^\top \;\longrightarrow\; J(T) \qquad (h \to 0),$$

where

$$J(T) = \frac{1}{\pi}\begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}\big(q''(T) + 2q(T)^3\big) \end{pmatrix}. \tag{3.1}$$

Gap 3.2. Taylor expand $\Phi(T \pm h)$ to the order needed for the $(1,1)$, off-diagonal, and $(2,2)$ entries after applying P_h; verify cancellation of singular powers of h; the $\frac{1}{12}$ coefficient and the $2q(T)^3$ cubic correction in (3.1) require an independent recheck. v1 prose to mine: the local kernel and jet normalization section.

3.2. Cross-block jet-limit.

Lemma 3.3 (Cross-block jet-limit)**.** *For $T_1 \neq T_2$, set $s = T_1 - T_2$, $\Delta = \Phi(T_1) - \Phi(T_2)$, $q_j = q(T_j)$, and let C_h be the 2×2 mixed block of K_Φ between the pairs $T_1 \pm h$ and $T_2 \pm h$. Then*

$$P_h\, C_h\, P_h^\top \;\longrightarrow\; \frac{1}{\pi} N_{12}(T_1, T_2) \qquad (h \to 0),$$

where

$$N_{12} = \begin{pmatrix} \dfrac{2\sin\Delta}{s} & \dfrac{\sin\Delta - q_2 s\cos\Delta}{s^2} \\ \dfrac{q_1 s\cos\Delta - \sin\Delta}{s^2} & \dfrac{(q_1+q_2)s\cos\Delta + (q_1q_2s^2 - 2)\sin\Delta}{2s^3} \end{pmatrix}. \tag{3.2}$$

Gap 3.4. Four-entry Taylor expansion in the two displacement variables; the sign convention on s and Δ controls whether the $(1,2)$ or $(2,1)$ entry carries q_1 or q_2 and must be audited before (3.2) is used downstream.

3.3. Same-point Gram positivity.

Lemma 3.5 (Same-point Gram positivity)**.** *Under (P1)–(P2) of Definition 2.1, the block $J(T)$ is positive definite on the retained packets at height T, and admits an absolute-exponent polynomial lower spectral bound*

$$\lambda_{\min}\big(J(T)\big) \geq Q^{-C_J}.$$

Gap 3.6. Reduce to a quadratic in q, q', q''; use (P2) for the lower bound on q and a derivative-of-zeta-phase upper bound for q''; uniformity in T on the retained set is the nontrivial part. v1 prose to mine: the curvature estimate subsection and the tail curvature bound section.

4. Finite-scale local blocks and whitening

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

The jet-limit identities of Section 3 are not directly usable: the comparison theorems of Section 8 requires the same blocks at finite separation $s > 0$ uniformly in s and the midpoint m. Whitening is the change of basis that absorbs the positivity of J into the normalization and produces a single dimensionless finite-scale packet block $\widehat{\Omega}_\zeta(s; m)$, which is the input to every comparison theorem from Section 5 onward.

4.1. **Finite same-point and mixed blocks.** Fix a midpoint m and separation $s > 0$; set $t_\pm = m \pm s/2$, $q_\pm = q(t_\pm)$, $\Delta_m(s) = \Phi(t_-) - \Phi(t_+)$.

Definition 4.1 (Finite same-point and mixed blocks)**.**

$$G_{m,\pm}(s) = \frac{1}{\pi}\begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}\big(q''(t_\pm) + 2q(t_\pm)^3\big) \end{pmatrix}, \tag{4.1}$$

$$N_m(s) = \frac{1}{\pi}\begin{pmatrix} -\dfrac{2\sin\Delta_m(s)}{s} & \dfrac{\sin\Delta_m(s) + q_+ s\cos\Delta_m(s)}{s^2} \\ -\dfrac{\sin\Delta_m(s) + q_- s\cos\Delta_m(s)}{s^2} & \dfrac{(q_- + q_+)s\cos\Delta_m(s) + (2 - q_- q_+ s^2)\sin\Delta_m(s)}{2s^3} \end{pmatrix}. \tag{4.2}$$

4.2. **Whitened finite block.**

Definition 4.2 (Whitened finite block)**.** Provided $G_{m,\pm}(s)$ are positive definite, set

$$\widehat{\Omega}_\zeta(s;m) := G_{m,-}(s)^{-1/2}\, N_m(s)\, G_{m,+}(s)^{-1/2}. \tag{4.3}$$

Lemma 4.3 (Whitening loss)**.** *On the retained packets there is an absolute exponent C_W such that the whitening map has operator-norm distortion at most Q^{C_W}; equivalently*

$$\big\|G_{m,\pm}(s)^{-1/2}\big\|_{\mathrm{op}} \;\le\; Q^{C_W}$$

uniformly in m and s on the retained set.

Gap 4.4. The diagonal of $G_{m,\pm}$ is bounded below by Lemma 3.5; bound the inverse via the explicit 2×2 formula; uniformity in m, s is the nontrivial part. v1 prose to mine: the corrected finite-s holomorphic whitening section.

5. Toy anomaly and transverse obstruction

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

The next three sections develop the transverse obstruction. The off-line toy phase Φ_{toy} defined below has an explicit transverse mode under the value/gauge projection Π_{trans} of Section 6; the actual zeta phase, under the same Π_{trans}, suppresses the same mode at a strictly larger polynomial exponent (Section 7). The transverse obstruction is the gap between these two bounds.

Definition 5.1 (Off-line toy phase)**.** For $\beta \in (\frac{1}{2}, 1)$ and a window scale Q, the toy off-line phase $\Phi_{\mathrm{toy}}(t;\beta,Q)$ is the model phase of a split pair at offset $\pm(2\beta-1)$ on a window of size $1/Q$, in the same coordinate and gauge as Definition 2.1.

Gap 5.2. Concrete formula for Φ_{toy} and identification of its q, q', q'' with elementary functions of $(2\beta-1)$ and the local coordinate. v1 prose to mine: the toy anomaly and transverse obstruction section.

Theorem 5.3 (Toy anomaly visibility)**.** *For every off-line distance $\beta - \frac{1}{2} \neq 0$ at scale Q, the toy whitened block $\widehat{\Omega}_{\mathrm{toy}}(s;m)$ satisfies*

$$\big\|\Pi_{\mathrm{trans}}\,\mathcal{A}_{\mathrm{toy}}\big(\widehat{\Omega}_{\mathrm{toy}}(s;m)\big)\big\| \;\ge\; Q^{-A_{\mathrm{toy}}}$$

on a positive polynomial-weight subset of midpoints m and separations s, where A_{toy} is an absolute exponent and $\mathcal{A}_{\mathrm{toy}}$ is the toy anomaly functional defined explicitly from the entries of $\widehat{\Omega}_{\mathrm{toy}}$.

Gap 5.4. Definition of $\mathcal{A}_{\mathrm{toy}}$ (a polynomial combination of the entries of $\widehat{\Omega}_{\mathrm{toy}}$ selected to vanish at $\beta = \frac{1}{2}$ but not for $\beta \neq \frac{1}{2}$) and the lower bound on retained (m, s). The bound is to be insensitive to the value channel and the residual gauge.

6. Transverse projection

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

The transverse projection Π_{trans} is the fixed value/gauge quotient under which the toy whitened block of Section 5 and the actual-zeta whitened block of Section 7 are compared. Its construction is the most sensitive structural decision in the rebuild: Π_{trans} is fixed here and is not to be re-chosen afterward.

Definition 6.1 (Transverse projection)**.** Let $\mathcal{V}_{\text{val}}$ denote the finite-dimensional subspace of the value/derivative space spanned by the value, gauge, and background directions to be quotiented out. A transverse projection Π_{trans} is an operator onto the orthogonal complement of $\mathcal{V}_{\text{val}}$ satisfying, on the retained packets:

(Π1) Π_{trans} annihilates $\mathcal{V}_{\text{val}}$ exactly;
(Π2) Π_{trans} is bounded with polynomial-in-Q loss;
(Π3) Π_{trans} is invariant under the residual gauge freedom left in Φ after Definition 2.1.

Gap 6.2. The choice of $\mathcal{V}_{\text{val}}$ must precede Theorem 5.3 and Hypotheses 7.1 and 7.3; it is not to be re-chosen after any has been invoked. v1 prose to mine: the leading value matrix and canonical calibration section, together with the local projective rigidity of the one-pair and multi-pair families section.

7. Actual-zeta transverse suppression

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

The actual zeta phase Φ defines two whitened blocks — the two-center block $\widehat{\Omega}_\zeta(s;m)$ and a four-center analogue $\widehat{\Omega}_\zeta^{(4)}$ for the symmetric quartet — on which the actual-zeta anomaly is suppressed at a strictly larger polynomial exponent than the corresponding toy lower bound of Theorem 5.3. This is the load-bearing asymmetry of the program: the off-line scenario forces the projective anomaly to be large; the actual zeta phase forces it to be small. The two-point and four-point comparisons of Section 8 consume different forms of this suppression, stated separately below.

7.1. **Two-point suppression.**

Hypothesis 7.1 (Two-point actual-zeta transverse suppression)**.** For the two-center packet associated with ρ and a symmetry partner, after the gauge-fixed normalization of Definition 2.1 and the projection Π_{trans} of Definition 6.1,

$$\left\|\Pi_{\text{trans}}\,\mathcal{A}_2\big(\widehat{\Omega}_\zeta(s;m)\big)\right\| \;\le\; Q^{-B_{\zeta,2}}$$

for an absolute exponent $B_{\zeta,2} > A_{\text{toy}}$ on the retained subfamily used in Theorem 5.3. Here $\mathcal{A}_2$ is the restriction of the toy anomaly functional of Theorem 5.3 to the actual-zeta two-center whitened block.

Gap 7.2. The two-point case of the structural gap. Consumed by Theorem 8.1. Admissible proof routes: (i) a standalone route from the local jet-Gram geometry alone; (ii) an energy-supply route consuming the source bound of Section 12 via Lemma 12.5. v1 prose to mine: the odd holomorphic transverse channel and corrected finite-s holomorphic whitening sections.

7.2. **Four-point suppression.**

Hypothesis 7.3 (Four-point actual-zeta transverse suppression)**.** For the four-center packet associated with the symmetry quartet $\{\rho,\bar\rho,1-\rho,1-\bar\rho\}$, under the four-point projection $\Pi^{(4)}_{\text{trans}}$ extending Definition 6.1,

$$\left\|\Pi^{(4)}_{\text{trans}}\,\mathcal{A}_4\big(\widehat{\Omega}^{(4)}_\zeta\big)\right\| \;\le\; Q^{-B_{\zeta,4}}$$

for an absolute exponent $B_{\zeta,4} > A^{(4)}_{\text{toy}}$ on the retained four-point subfamily. Here $\mathcal{A}_4$ is the four-point analogue of the toy anomaly functional and $A^{(4)}_{\text{toy}}$ is the four-point toy visibility exponent.

Gap 7.4. The four-point case of the structural gap. Consumed by Theorem 8.4. Same admissible proof routes as Hypothesis 7.1. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families section, four-point variant.

Remark 7.5 (Standalone and energy-supply routes)**.** Hypotheses 7.1 and 7.3 are stated independently of the technology of their proofs. A purely local proof would use only the jet-Gram and quotient geometry of Sections 3–6; an energy-supply route may use the source-energy module of Section 12. Forward references from Section 8 indicate which hypothesis is invoked, not which route proves it.

8. Local projective rigidity

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

Theorem 5.3 provides the toy lower bound and Hypotheses 7.1 and 7.3 provide the matching actual-zeta upper bounds on the transverse anomaly under Π_{trans} of Definition 6.1. The present section assembles the comparison. On the two-point family attached to ρ and a single symmetry partner, and on the four-point family of the full symmetric quartet $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$, the exponent inequality $A_{\mathrm{toy}} < B_\zeta$ (resp. its four-point analogue $A^{(4)}_{\mathrm{toy}} < B^{(4)}_\zeta$) is incompatible with the existence of the configuration. The two cases share the comparison mechanism but differ in the underlying jet-Gram object: the two-point case uses the cross-block N_{12} of Lemma 3.3; the four-point case uses a four-center analogue $\widehat{\Omega}^{(4)}_\zeta$ and a four-point extension $\Pi^{(4)}_{\mathrm{trans}}$ of the projection of Definition 6.1. Either case suffices for the aggregation step in Section 10.

8.1. **Two-point comparison.** The two-point comparison takes ρ and one symmetry partner (typically $\bar{\rho}$ or $1-\rho$) and reads off the joint contribution to $N_m(s)$ at matched midpoint and separation. The exponent inequality $A_{\mathrm{toy}} < B_\zeta$ excludes the pair.

Theorem 8.1 (Two-point rigidity)**.** *Let $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ and $|\gamma|$ sufficiently large, and let (T_1, T_2) be the height pair attached to ρ and one of its symmetry partners. Under Lemmas 3.1, 3.3, 4.3, Theorem 5.3, and Hypothesis 7.1,*

$$A_{\mathrm{toy}} < B_{\zeta,2} \qquad \Longrightarrow \qquad \rho \notin \{\zeta = 0\}.$$

Gap 8.2. Compose Theorem 5.3 (lower bound) with Hypothesis 7.1 (upper bound) at matched (m, s); show the exponent gap survives the polynomial whitening loss (Lemma 4.3) and the polynomial Gram lower bound (Lemma 3.5). The proof is short — arithmetic on absolute exponents — once the inputs are in place.

Remark 8.3 (Energy-supply route, if invoked)**.** If Hypothesis 7.1 cannot be discharged from the local jet-Gram geometry alone for the two-point case, the source-energy lower bound Hypothesis 12.3 of Section 12, transferred via Lemma 12.5, supplies the missing polynomial-in-Q room. Theorem 8.1 is then conditional on that source as well as on the local inputs.

8.2. **Four-point comparison in the symmetric quartet.** The four-point comparison uses the full symmetry packet $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$. The four-center jet-Gram block carries strictly more rigidity than the two-point cross-block: the toy visibility lower bound holds in a refined quotient invisible to the two-point comparison, and the matching suppression hypothesis is correspondingly stronger. The four-point theorem is stated independently of Theorem 8.1.

Theorem 8.4 (Four-point rigidity)**.** *Let ρ be a hypothetical off-line zero and $\{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}$ its symmetry quartet. Under the four-point analogue of Theorem 5.3 and Hypothesis 7.3,*

$$A^{(4)}_{\mathrm{toy}} < B_{\zeta,4} \qquad \Longrightarrow \qquad \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\} \not\subset \{\zeta = 0\}.$$

Gap 8.5. State and prove the four-center analogue $\widehat{\Omega}^{(4)}_\zeta$ of $\widehat{\Omega}_\zeta$; the four-point projection $\Pi^{(4)}_{\mathrm{trans}}$; and the four-point analogue of Theorem 5.3 matching Hypothesis 7.3. Assemble the exponent inequality. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families section, together with the corrected fixed-target quotient package.

Remark 8.6 (Energy-supply route, if invoked)**.** The four-point analogue of Remark 8.3 applies: if Hypothesis 7.3 cannot be proved from local geometry alone, Hypothesis 12.3 of Section 12 supplies the missing room via the four-point branch of Lemma 12.5, separating $A_{\mathrm{toy}}^{(4)}$ from $B_{\zeta,4}$.

9. N-point odd-moment cancellation

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

The exponent inequality of Theorems 8.1 and 8.4 is load-bearing, but its inputs include finite remainders from the Taylor expansion of Φ around midpoints. By the symmetry of the packet measure, all odd moments of the relevant expansion vanish; this is the mechanism that holds the remainders below $\min(Q^{-A_{\mathrm{toy}}}, Q^{-B_\zeta})$ and lets the comparison close uniformly in the height.

Lemma 9.1 (N-point odd-moment cancellation)**.** *Let $\widehat{\mu}_{N,Q}$ be the symmetric N-point packet measure attached to height T and scale Q. For every $r \geq 0$,*

$$\partial_z^{2r+1}\widehat{\mu}_{N,Q}(0) = 0. \tag{9.1}$$

The same identity holds with $\widehat{\mu}_{N,Q}$ replaced by its convolution with any even bump $\widehat{\phi}$: for every $r \geq 0$,

$$\partial_z^{2r+1}\big(\widehat{\phi}(\eta z)\widehat{\mu}_{N,Q}(z)\big)\big|_{z=0} = 0. \tag{9.2}$$

Gap 9.2. Pure parity argument; v1 carries the full proof in the N-point odd-moment cancellation section. Migration is straightforward provided the symmetry packet of (9.1) matches Definition 2.1.

10. Aggregation and zero-free strip

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

The two-point and four-point rigidity theorems exclude any single off-line zero above height T_0; the upgrade to a zero-free strip proceeds by aggregating along the dyadic height ladder, exploiting the polynomial nature of every loss in Lemmas 4.3, 3.5, Theorem 5.3, and Hypotheses 7.1 and 7.3.

Theorem 10.1 (Zero-free strip)**.** *There is $T_0 > 0$ such that $\zeta(\beta + i\gamma) \neq 0$ for every $\beta \neq \frac{1}{2}$ and $|\gamma| \geq T_0$.*

Gap 10.2. Aggregate Theorem 8.1 or Theorem 8.4 across the dyadic ladder; the polynomial losses combine into a single polynomial-in-$Q(\gamma)$ factor, and the exponent inequality $A_{\mathrm{toy}} < B_\zeta$ carries through uniformly.

11. The Riemann hypothesis

paper: [**LIVE**] v1: [**n/a**] referee: [**not-started**] python: [**n/a**] lean: [**not-started**]

Theorem 11.1 (Riemann hypothesis)**.** *Every nontrivial zero of ζ satisfies* $\operatorname{Re} s = \frac{1}{2}$.

Proof. By Theorem 10.1, no off-line zero can lie at height $|\gamma| \geq T_0$; the standard low-height numerical verification of RH (see, e.g., [3, 5]) covers the range $|\gamma| < T_0$. □

12. Source-energy supply (optional)

paper: [**LIVE**] v1: [**not-started**] referee: [**not-started**] python: [**not-started**] lean: [**not-started**]

This section is self-contained: its sole role is to supply a single source-energy lower bound that Theorems 8.1 and 8.4 may invoke when their standalone forms in Section 8 are insufficient. Nothing earlier in the paper depends on this section.

12.1. **Canonical source curvature.**

Definition 12.1 (Canonical source curvature)**.** The canonical source curvature of ζ on the chart of Definition 2.1 is

$$q''_{\mathrm{can}} := \partial_t^3 Z(t), \tag{12.1}$$

where $Z(t)$ is the real critical-line scalar of ζ in the same chart.

Gap 12.2. Identification of q''_{can} with $\partial_t^3 Z$ is a convention; chart-regularity and zero-separation hypotheses of Definition 2.1 carry over verbatim. v1 prose to mine: the upstream Gate-1 arithmetic inputs section.

12.2. **Source-energy bound.**

Hypothesis 12.3 (Source-energy lower bound)**.** On retained packet intervals $I \subset I_T$, the canonical source curvature satisfies

$$\int_I |q''_{\mathrm{can}}(m)|^2 \, dm \;\geq\; Q^{-A_E} \, |I| \tag{12.2}$$

on a positive polynomial-weight subfamily, for an absolute exponent A_E.

Gap 12.4. polynomial lower bound on $\partial_t^3 Z$ across the retained packet family. This is an analytic estimate on ζ and not on the local jet-Gram geometry; it is what makes Section 12 optional. v1 prose to mine: the Hardy-curvature mass-good packet source block.

12.3. **Energy-to-suppression transfer.**

Lemma 12.5 (Source energy to transverse suppression)**.** *Hypothesis 12.3, after carrier decomposition and packet trace normalization, implies one of Hypothesis 7.1 (with exponent $B_{\zeta,2} = B_{\zeta,2}(A_E) > A_{\mathrm{toy}}$) or Hypothesis 7.3 (with exponent $B_{\zeta,4} = B_{\zeta,4}(A_E) > A^{(4)}_{\mathrm{toy}}$), according to which carrier transfer is established into the corresponding transverse channel of $\widehat{\Omega}_\zeta(s;m)$ or $\widehat{\Omega}^{(4)}_\zeta$.*

Gap 12.6. Carrier decomposition and packet trace step taking $\int |q''_{\mathrm{can}}|^2$ on the source side into a lower bound on $\Pi_{\mathrm{trans}}\widehat{\Omega}_\zeta$ (two-point case) or $\Pi^{(4)}_{\mathrm{trans}}\widehat{\Omega}^{(4)}_\zeta$ (four-point case). v1 prose to mine: the certified downstream status after the canonical source-graded closure section, and the carrier-corrected source bridge package.

Appendix A. Archive map

paper: [**DIAGNOSTIC**] v1: [n/a] referee: [n/a] python: [n/a] lean: [n/a]

The previous large draft is preserved in place at `rh/proof_of_rh.tex` and is treated as a quarry from which prose and computations may be migrated, after restatement and refereeing, into the present file. Pointers to relevant v1 sections are given inline at each `wip` mark above; the table below summarizes the principal mining correspondences.

Rebuild section	**v1 source (in `proof_of_rh.tex`)**
§2	Local kernel and jet normalization.
§3	Finite-s local blocks; jet-limit local blocks.
§4	Corrected finite-s holomorphic whitening.
§5	Toy anomaly and transverse obstruction.
§6	Leading value matrix and canonical calibration.
§7	Odd holomorphic transverse channel; corrected fixed-target quotient package.
§8	Local projective rigidity of the one-pair and multi-pair families; finite coefficient table contracts.
§9	N-point odd-moment cancellation.
§10, §11	Master conditional theorem; introduction.
§12	Hardy-curvature mass-good packet source block; carrier-corrected source bridge package; upstream Gate-1 arithmetic inputs.

The v1 sections "Endgame architecture and remaining propagation problem outside the clean mixed four-point case," "Downstream local closure package: Gate 2, wall escape, and carrier rows," "Unified finite-survival architecture," "Current proof state after the local package audits," "How the older framing maps to the new framing," and "No-go and audit ledger" are deliberately not migrated; their content is route history and process bookkeeping rather than active proof material.

References

[1] E. Bombieri, *Problems of the millennium: the Riemann hypothesis*, Clay Math. Inst. (2000).

[2] J. B. Conrey, *The Riemann hypothesis*, Notices Amer. Math. Soc. **50** (2003), 341–353.

[3] H. M. Edwards, *Riemann's Zeta Function*, Dover Publications, Mineola, NY, 2001.

[4] B. Riemann, *Über die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Berl. Akad. (1859), 671–680.

[5] E. C. Titchmarsh (rev. D. R. Heath-Brown), *The Theory of the Riemann Zeta-function*, Oxford University Press, 1986.